

Ad Launch - Project Instructions

Overview

Ad Launch is a lead-gen platform that takes a business URL, performs deep research via Tombstone OS (FastAPI backend), and generates visual Facebook ads, SEO audits, Google Ads copy, website concepts, budget recommendations, and 90-day posting plans.

Architecture

- **Frontend:** Next.js 14 (App Router) with Tailwind CSS, deployed at `ad-launch-1nfyr8.abacusai.app`
- **Backend Pipeline:** Tombstone OS (FastAPI on Render) — multi-agent system for research + creative generation
- **Database:** PostgreSQL (Prisma ORM)
- **Auth:** NextAuth with credentials (business email only)
- **CRM:** GoHighLevel (GHL) for contact management + confirmation emails
- **Image Storage:** S3 (AWS) for GPT-5.1 generated ad images; R2 (Cloudflare) for legacy Tombstone artifacts

Ad Generation Pipeline (GPT-5.1 Upgrade - April 2026)

- **Previous:** Andy Warhol (FAL Flux) → Claude Hopkins (text overlay) → basic stock photo with text
- **Current:** After Tombstone completes research + creative, `mission-status/route.ts` calls `generateAllAdImages()` which uses **GPT-5.1 “Designer Brief”** prompt pattern to generate professional multi-section Facebook ad compositions
- **Prompt Strategy:** Agency brief style (“You are a senior graphic designer...”) with structured sections: CLIENT, CREATIVE BRIEF, DESIGN DIRECTION
- **Fallback:** If GPT-5.1 generation fails, falls back to original Tombstone FAL images
- **Storage:** GPT-5.1 images are uploaded to S3 as public PNGs, URL stored in `Ad.imageUrl`
- **Edit Ads:** `/api/edit-ad` also uses GPT-5.1 instead of `gpt_image15`

Key Models Used

- `gpt-5.1` — Ad image generation (Designer Brief pattern), ad editing
- `gpt-4.1` — Complex text generation (concept sites, etc.)
- `gpt-4.1-mini` — Search, lighter text tasks

Image URL Resolution

- S3 URLs (`.s3.amazonaws.com`) → passed through directly (GPT-5.1 generated)
- R2 keys → resolved via Tombstone `/artifacts/resolve` endpoint (legacy)
- Data URLs → passed through directly

Test Lab

- `/test-ads` page with `/api/test-ad-gen` endpoint for A/B testing different models and prompt strategies
- Test images saved in `/public/test-results/`

Visual Style

- Primary: #2563EB (blue), Dark: #0F172A, Light: #F8FAFC
- Clean, modern UI with generous whitespace
- Watermarks on free tier ads (“Ad Launch” diagonal text)

Important Files

- `lib/generate-ad-image.ts` — GPT-5.1 ad generation + S3 upload
- `lib/aws-config.ts` — S3 client configuration
- `lib/tombstone.ts` — Tombstone OS API integration
- `lib/seo-audit.ts` — Live SEO audit fallback
- `app/api/mission-status/route.ts` — Main pipeline orchestration
- `app/api/edit-ad/route.ts` — Ad editing with GPT-5.1

RSS Intelligence System (Phase 3 complete — April 2026)\n- Phase 3: 190 RSS feeds discovered across 5 pilot states (CO=51, TX=71, FL=29, NC=14, MT=25). `lib/rss/discovery.ts` (HTML link + path probe), `lib/rss/source-classifier.ts`, `data/pilot-seed-sources.json` (118 curated sources), `scripts/discover-feeds.ts`. Schema added `discoveryMethod/feedFormat/pilotState` to `RssFeed`.

- Phase 2: 31,273 delivery-point ZIPs (user’s whitelist, PO Box excluded), 49 states, 3,095 counties, 26,609 cities. `lib/rss/geo-lookup.ts` provides radius/ZIP-list/county/state/city trade area queries with Haversine. Import script: `scripts/import-geo-data.ts`.
- **Schema:** 10 new models added to Prisma — `ContentPolicy`, `GeoState/County/City/Zip/CityZip` (hierarchy), `RssFeed`, `RssItem`, `FeedGeo` (M:N with confidence), `FeedAudit`, `ItemAudit`
- **Content Policy Engine:** `lib/rss/content-policy.ts` — multi-layer keyword filter (sexual/adult hard-block, political/opinion hard-block, 6 soft-filter categories, 11 allow categories)
- **Types:** `lib/rss/types.ts` — shared types for source taxonomy, policy actions, trade area query contract
- **Seed:** `scripts/seed-content-policies.ts` + policies embedded in `scripts/seed.ts` (19 policies)
- **Tombstone Integration:** `RssItem` has `usedInPost`, `usedAt`, `postReference` fields for creative workflow consumption; `RssFeed` has `lastAgentRun` / `lastAgentRunAt` for agent tracking
- **Build Plan:** `docs/RSS_INTELLIGENCE_BUILD_PLAN.md` — full 10-phase plan
- **Next phase:** Phase 2 (Geography reference layer — import US ZIP/city/county data)